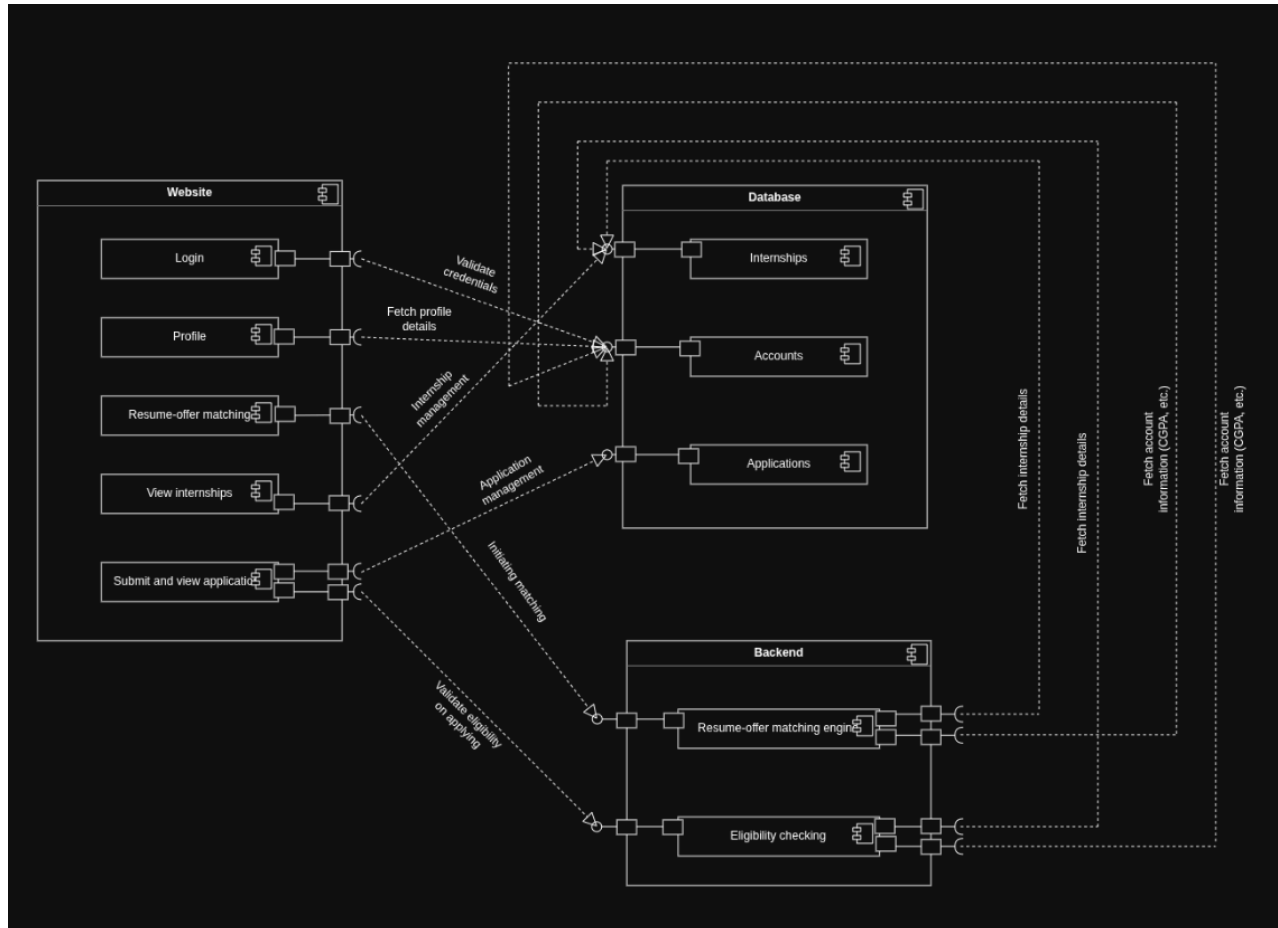


Problem Statement #3 - Campus Placement & Internship Pipeline

Component diagram:



Architecture justification:

We chose a microservice architecture for the campus placement and internship pipeline. This is because this architecture is easily extensible and extra features such as conduction of OAs on the platform itself can be added later. It is also easy to debug due to fault isolation because the microservices are independent.

A security advantage of a microservice architecture is that due to the fact that the microservices are all independent, a breach in one does not compromise a large part of the entire system.

A performance benefit of microservices is that bottlenecks are reduced, since there are many different connection points, so popular services will not slow down the less popular ones by hogging the server/connections.

Connection justifications:

1. Login to accounts: You need the account's credentials to log in
2. Profile to accounts: The profile details such as name, CGPA, etc. are stored along with the account information in the database.
3. Resume-offer matching to resume-offer matching engine: The user initiates the matching from the frontend, and the request is then sent to the backend for processing.
4. View internships to internships: For the user to view the currently being offered internships they must fetch the internship details from the database.
5. Submit and view applications to applications: For the user to view their applications, they must fetch the application details of their applications from the database.
6. Submit and view applications to eligibility check: For the user to successfully submit an application, they must be eligible for it, which will be checked by the backend before submission.
7. Resume-offer matching engine to internships: For the system to match a resume to its best fitting internship, it must be able to view all internship details.
8. Resume-offer matching engine to internships: For the system to match a resume to its best fitting internship, it must be able to view the student's uploaded resume which will be attached to his account. An alternative to this would be just asking them to upload a resume while initiating the matching, and in that case this connection would not be required.
9. Eligibility check to internships: For the system to validate whether a student is eligible for a certain internship, it must check the internship's details first.
10. Eligibility check to accounts: For the system to validate whether a student is eligible for a certain internship, it must also be able to access the student's credentials such as CGPA, which is stored in accounts.